NSW Education Standards Authority

Centre Number

Student Number

2026 **HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

Total marks: 100

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6–19)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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1 The polynomial $f(x) = x^3 - 10x^2 + 34x - 40$ has roots $3 + i$, α and β.

What are the values of α and β?

A. $3 - i, 4$

B. $3 + i, 4$

C. $3 - i, -16$

D. $3 + i, -16$

2 What is the length of the vector $-2\underset{\sim}{i} + 3\underset{\sim}{j} - 6\underset{\sim}{k}$?

A. 5

B. 7

C. 11

D. 49

3 Which equation describes a sphere of radius 9, centred at $(-1, 3, 0)$?

A. $x^2 + 2x + y^2 + 6y + z^2 = 71$

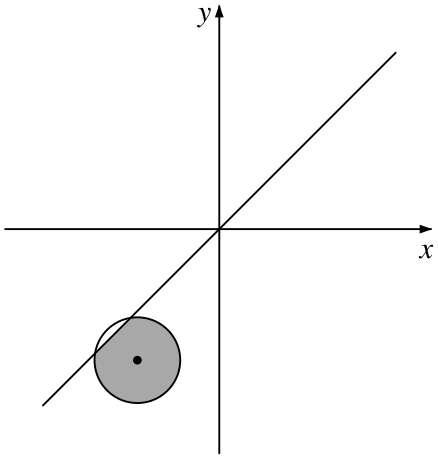
B. $x^2 - 2x + y^2 - 6y + z^2 = 71$

C. $x^2 - 2x + y^2 + 6y + z^2 = 71$

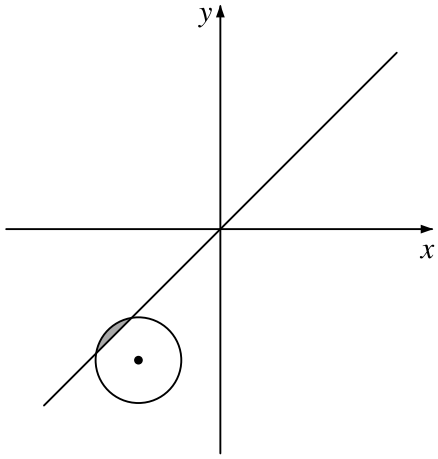
D. $x^2 + 2x + y^2 - 6y + z^2 = 71$

4 Which of the following diagrams represents the region satisfied by the inequalities $|z - 2 - 3i| \leq 1$ and $\text{Re}(z) \geq \text{Im}(z)$?

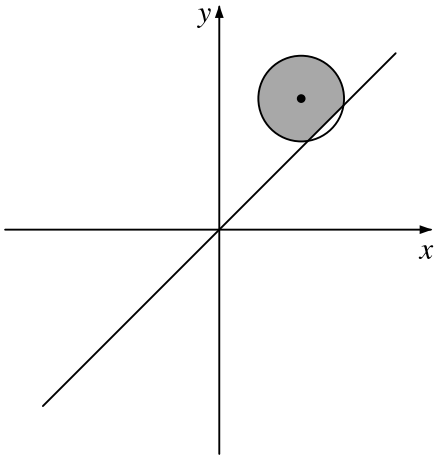
A.



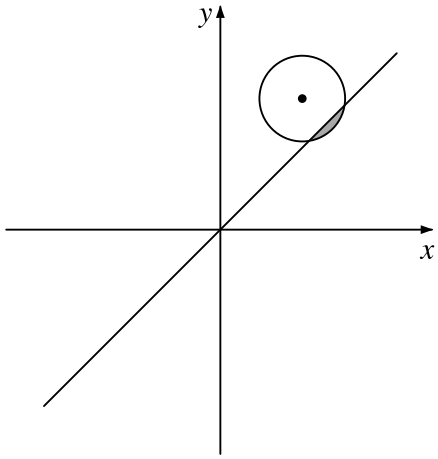
B.



C.



D.



5 Let z and w be complex numbers, such that $\frac{w}{z} = 3 + 4i$.

If $\theta = \arg(w) - \arg(z)$, what is the value of $\sin\theta$?

A. $\frac{3}{5}$

B. $\frac{4}{5}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

6 Consider the statement:

$$\forall p > 0 \quad \exists q > p \quad \text{such that} \quad q^2 = p.$$

Which of the following is the negation of the statement?

A. $\exists p \leq 0 \quad \text{such that} \quad q^2 \neq p \quad \forall q \leq p$

B. $\exists p \leq 0 \quad \text{such that} \quad q^2 = p \quad \forall q \leq p$

C. $\exists p > 0 \quad \text{such that} \quad q^2 \neq p \quad \forall q > p$

D. $\exists p > 0 \quad \text{such that} \quad q^2 = p \quad \forall q > p$

7 Which of the following could be a primitive function of

$$\frac{7}{(x+1)(x-6)^2}$$

where A, B and C are constants?

A. $A \log_e|x+1| + \dfrac{B}{x-6} + C$

B. $A \log_e|x+1| + \dfrac{B}{(x-6)^3} + C$

C. $A \log_e|x+1| + B \log_e|x-6| + \dfrac{C}{x-6}$

D. $A \log_e|x+1| + B \log_e|x-6| + \dfrac{C}{(x-6)^3}$

8 What is the area of the triangle with vertices at $2z$, $-iz$ and $2z - iz$, where z is a non-zero complex number?

A. $\dfrac{1}{2}|z|^2$

B. $|z|^2$

C. $\dfrac{\sqrt{5}}{4}|z|^2$

D. $\sqrt{5}|z|^2$

9 Consider the lines:

$$\ell_1 : \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad \text{and} \quad \ell_2 : \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 2 \\ -4 \end{pmatrix}$$

where λ and μ are parameters.

Which of the following is true?

A. The lines are intersecting and parallel.

B. The lines are non-intersecting and parallel.

C. The lines are intersecting and perpendicular.

D. The lines are non-intersecting and perpendicular.

10 Consider the statement:

If the second derivative of a function is zero at $x = 0$
then the function has a point of inflection at $x = 0$.

Which of the following is a counter-example that proves the statement is false?

A. $f(x) = x^2 + 2$

B. $f(x) = x^3 + 1$

C. $f(x) = x^5 + x^4$

D. $f(x) = x^6 + x^5$

Section II

90 marks
Attempt Questions 11–16
Allow about 2 hours and 45 minutes for this section

Answer the questions in the Section II Writing Booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Answer Question 11 on pages 3–6 of the Writing Booklet.

(a) Write the contrapositive of the statement: **1**

If a^3 is even then a is even.

(b) If $z = 3 + 2i$ and $w = -1 - 5i$, find:

(i) $3z + w$ **1**

(ii) $2\bar{z}w$. **1**

(c) Let P_1 be the point $(1,0,0)$ and P_2 be the point $\left(\frac{1}{\sqrt{10}}, 0, \frac{3}{\sqrt{10}}\right)$. **2**

If θ is the angle between the position vectors $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$, find $\tan^2\theta$.

(d) Find the two square roots of $11 + 60i$ in the form $x + iy$, where x and y are real numbers. **2**

(e) Given $z = re^{i\theta}$, prove $z\bar{z} = |z|^2$. **2**

Question 11 continues on page 7

(f) Evaluate $\int_{\frac{3}{2}}^{3} \frac{1}{\sqrt{6x - x^2}}\, dx$. **3**

(g) If $z = a + ai$ where a is a positive real number, find:

(i) $\left|z^5\right|$ in terms of a **1**

(ii) the principal argument of z^5. **2**

End of Question 11

Please turn over

Question 12 (15 marks) Answer Question 12 on pages 8–11 of the Writing Booklet.

(a) (i) Given $e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$, show that **3**

$$\left(\frac{e^{i\theta} - e^{-i\theta}}{2i}\right)^4 = \frac{1}{8}(\cos 4\theta - 4\cos 2\theta + 3).$$

(ii) Hence find $\int \sin^4 \theta \, d\theta$. **2**

(b) Find a polynomial with real coefficients and zeroes 2, $1 + 6i$ and $3 - 4i$. **2**

Give your answer in factored form with linear and quadratic factors with real coefficients.

(c) It is known that **2**

$$\frac{x+y}{2} \geq \sqrt{xy} \qquad \text{(Do NOT prove this.)}$$

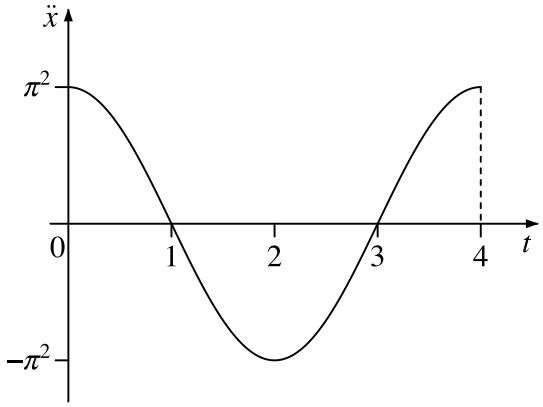
for all positive real numbers x, y.

Prove that if a, b and c are positive real numbers, then

$$(a + 7b)(b + 2c)(c + 14a) \geq 112abc.$$

Question 12 continues on page 9

(d) The acceleration $\ddot{x}$ of a particle moving in simple harmonic motion is given by the following graph. **3**



If the particle is initially at rest at $x = -2$, find an equation for its displacement in terms of time t.

(e) Find $\displaystyle\int \frac{e^x + e^{2x}}{1 + e^{2x}}\, dx$. **3**

End of Question 12

Question 13 (14 marks) Answer Question 13 on pages 13–16 of the Writing Booklet.

(a) By considering the pattern below, prove that any positive multiple of 4 can be written as a difference of squares. **2**

$$\begin{aligned} 4 &= 2^2 - 0^2 \\ 8 &= 3^2 - 1^2 \\ 12 &= 4^2 - 2^2 \\ 16 &= 5^2 - 3^2 \\ &\vdots \end{aligned}$$

(b) A particle moving in simple harmonic motion has speed v m s^{-1}, given by **2**

$$v^2 = -2x^2 + 8x + 8$$

where x is the displacement from the origin in metres.

Find the distance that the particle travels during one period of its motion.

(c) Find $\displaystyle\int \frac{d\theta}{5 + 3\cos\theta - 2\sin\theta}$. **3**

(d) A stone of mass 0.8 kg is projected vertically upwards from the ground and experiences air resistance with magnitude $0.08v^2$ newtons, where v m s^{-1} is the velocity of the stone. Let the acceleration due to gravity be 10 m s^{-2}. **3**

Given the maximum height reached by the stone is $5\ln\left(\frac{676}{100}\right)$ m, find the speed of the stone on its return to the point of projection. Give your answer correct to 2 decimal places.

(e) (i) Prove $\sqrt{6}$ is irrational. **2**

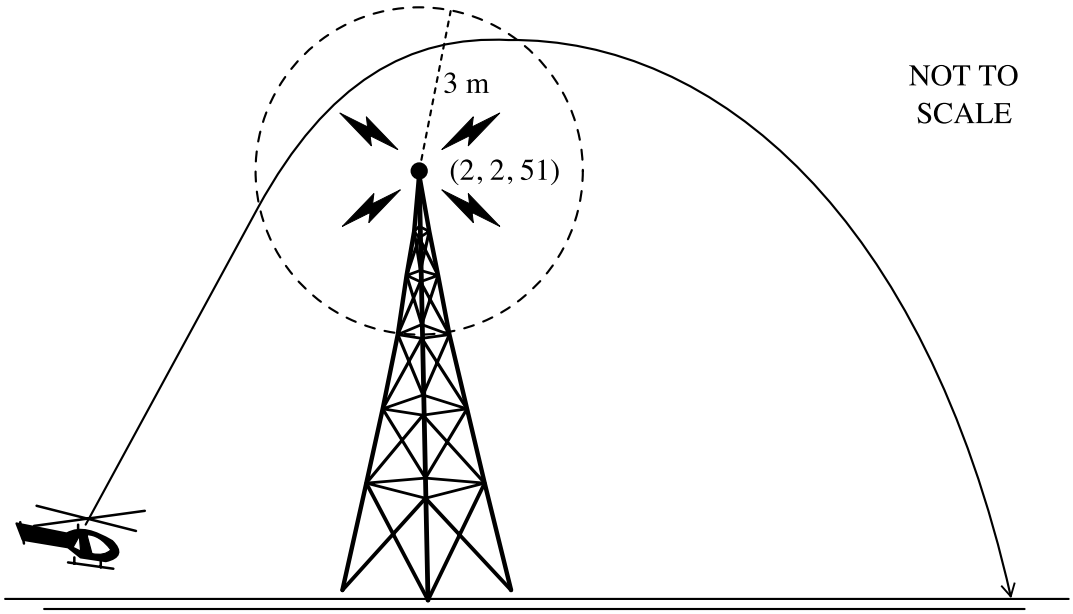
(ii) Hence, or otherwise, prove that $\sqrt{3} - \sqrt{2}$ is irrational. **2**

Question 14 (15 marks) Answer Question 14 on pages 18–21 of the Writing Booklet.

(a) A remote-controlled helicopter moves along the path **4**

$$\underset{\sim}{r}(t) = \left(5 - \frac{t}{4}\right)\underset{\sim}{i} + 4\underset{\sim}{j} + \left(51 + \frac{t}{4}\right)\underset{\sim}{k}$$

where t is the time in seconds, until it comes within 3 m of a radio transmitter located at $(2, 2, 51)$, after which the helicopter loses all power and begins to fall to the ground.

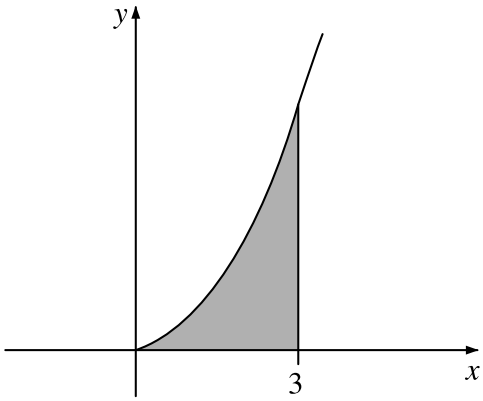


Let the acceleration due to gravity be 10 m s^{-2}.

From the time it loses power, how long does it take for the helicopter to reach the ground?

Question 14 continues on page 12

(b) Consider the region bounded by the curve $y = x\sqrt{e^x}$, the x-axis, and the line $x = 3$. **3**

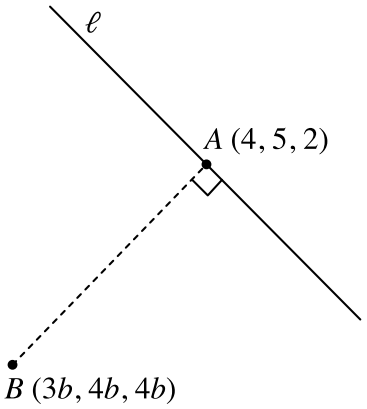


NOT TO SCALE

Find the volume of the solid of revolution formed when the region is rotated about the x-axis.

Question 14 continues on page 13

(c) The point $A(4,5,2)$ lies on the line $\ell : \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 2 \\ -3 \end{pmatrix}$ where λ is a parameter.



B is a point with position vector $\begin{pmatrix} 3b \\ 4b \\ 4b \end{pmatrix}$, where b is a constant. The vector $\overrightarrow{BA}$ is perpendicular to ℓ.

(i) Show that $b = 2$. **1**

(ii) The point C lies on ℓ such that $\angle ACB = \frac{\pi}{4}$. **2**

Find the two possible position vectors for C.

Question 14 continues on page 14

(d) A particle of mass m kg is launched and moves horizontally in a straight line with initial velocity u m s^{-1}, $u > 0$. The particle is resisted by a force with magnitude $mk + mv^2$ newtons, where k is a positive constant and v m s^{-1} is the velocity of the particle at time t seconds.

(i) Show that **2**

$$t = \frac{1}{\sqrt{k}}\left(\tan^{-1}\frac{u}{\sqrt{k}} - \tan^{-1}\frac{v}{\sqrt{k}}\right).$$

(ii) Hence, find the possible values of u such that the particle is moving in the negative direction when $t = \dfrac{\pi}{4\sqrt{k}}$ seconds. **3**

End of Question 14

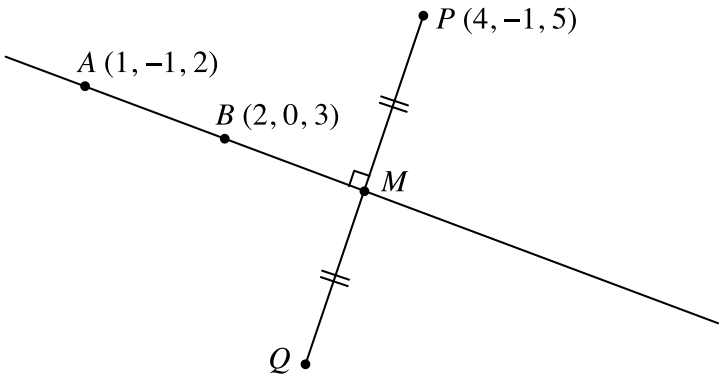
Question 15 (15 marks) Answer Question 15 on pages 23–26 of the Writing Booklet.

(a) Prove by mathematical induction for integers $n > 1$ that **4**

$$\frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{2n-1} + \frac{1}{2n} > \frac{13}{24}.$$

(b) Consider the line through the points $A(1,-1,2)$ and $B(2,0,3)$. **4**

The point M lies on the line and is the midpoint of points $P(4,-1,5)$ and Q.



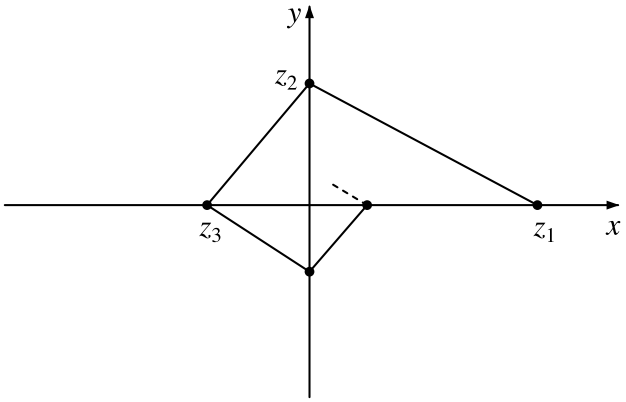
Find the coordinates of Q.

Question 15 continues on page 16

(c) Consider the sequence defined by **3**

$$z_1 = 8 \quad \text{and} \quad z_{n+1} = \frac{3}{4} i z_n$$

as shown.



NOT TO SCALE

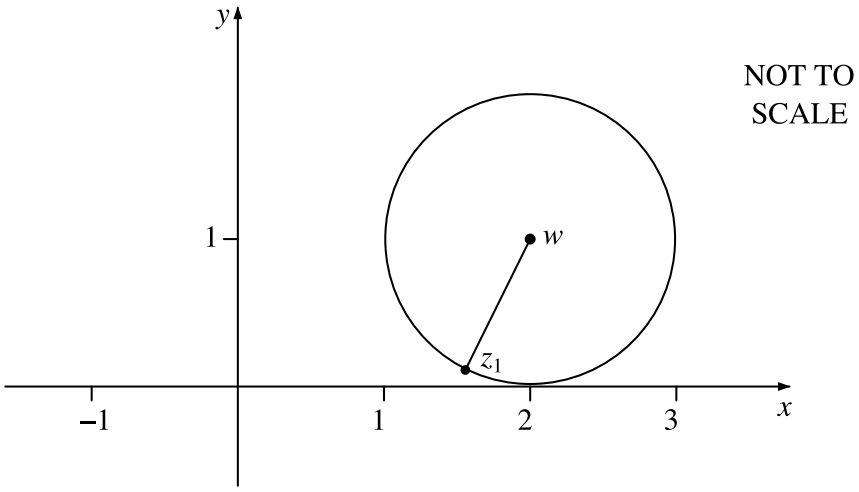
Find the sum of the lengths

$$|z_2 - z_1| + |z_3 - z_2| + |z_4 - z_3| + \ldots.$$

Question 15 continues on page 17

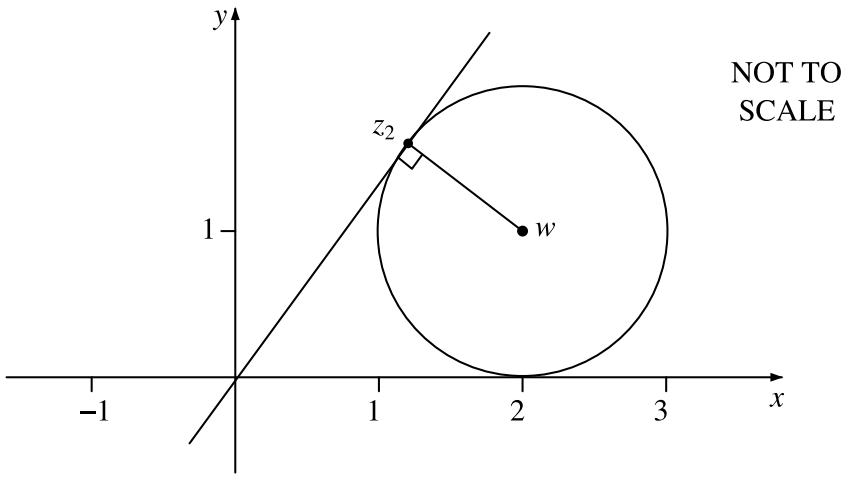
(d) Let $w = 2 + i$.

(i) Let z_1 be a complex number on the circle of radius 1, centred at w. **2**



If $\text{Arg}(z_1 - w) = -\dfrac{2\pi}{3}$, find z_1 in Cartesian form.

(ii) Let z_2 be a complex number on the circle of radius 1, centred at w. **2**



If the tangent to the circle at z_2 passes through the origin, find z_2 in Cartesian form.

End of Question 15

Question 16 (16 marks) Answer Question 16 on pages 28–31 of the Writing Booklet.

(a) Suppose z is a complex number on the unit circle such that **4**

$$\text{Im}\left(z^3 + z^2 + z\right) = 0.$$

Find all possible values of z.

(b) Let $I_n = \displaystyle\int_0^1 \frac{dx}{\left(x^2 + 1\right)^n}$ for integers $n \geq 1$.

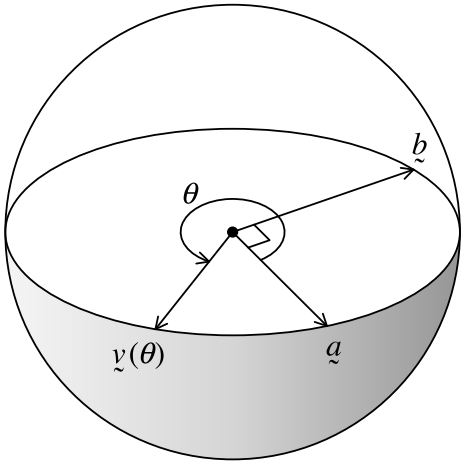
(i) Show that $I_n - I_{n-1} = \displaystyle\int_0^1 \frac{-x^2}{\left(x^2 + 1\right)^n}\, dx$. **1**

(ii) Hence evaluate I_2. **3**

Question 16 continues on page 19

(c) Suppose A and B are points on a sphere with radius 1 centred at the origin O. If $\underset{\sim}{a} = \overrightarrow{OA}$ is perpendicular to $\underset{\sim}{b} = \overrightarrow{OB}$, then the circle centred at O which passes through A and B has vector equation

$$\underset{\sim}{v}(\theta) = \cos\theta\, \underset{\sim}{a} + \sin\theta\, \underset{\sim}{b}, \quad 0 \le \theta < 2\pi. \qquad \text{(Do NOT prove this.)}$$



(i) Show that $|\underset{\sim}{v}(\theta)|^2 = 1$. **2**

(ii) Let C_1 and C_2 be circles with radius 1 and with vector equations **2**

$$C_1: \quad \cos\alpha \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \sin\alpha \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}, \quad 0 \le \alpha < 2\pi$$

$$C_2: \quad \cos\theta \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \sin\theta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad 0 \le \theta < 2\pi.$$

Find the points of intersection of C_1 and C_2.

(iii) Consider the circle with centre O which passes through the points $A(0,1,0)$ and $P\left(\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{6}}, -\frac{1}{2\sqrt{3}}\right)$. **4**

By projecting $\overrightarrow{OP}$ onto $\overrightarrow{OA}$, or otherwise, find a vector equation for the circle.

End of paper

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2026 **HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}\left[2a + (n - 1)d\right] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

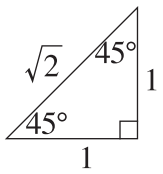
$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

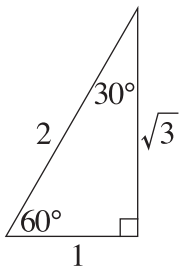
$$A = \frac{1}{2}ab\sin C$$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \ \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \ \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \ \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

If $t = \tan\frac{A}{2}$ then $\sin A = \frac{2t}{1 + t^2}$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A\cos B = \frac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$$

$$\sin A\sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$$

$$\sin A\cos B = \frac{1}{2}\left[\sin(A + B) + \sin(A - B)\right]$$

$$\cos A\sin B = \frac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

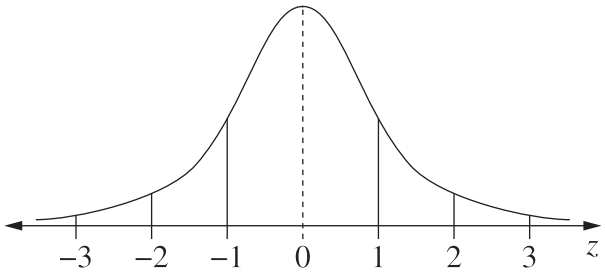
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E\left[(X - \mu)^2\right] = E\left(X^2\right) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x)\,dx$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \text{Bin}(n, p)$$

$$\Rightarrow \ P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \ x = 0, 1, \ldots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\frac{dy}{dx} = n f'(x)\left[f(x)\right]^{n-1}$
$y = uv$	$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - \left[f(x)\right]^2}}$
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - \left[f(x)\right]^2}}$
$y = \tan^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{1 + \left[f(x)\right]^2}$

Integral Calculus

$$\int f'(x)\left[f(x)\right]^n dx = \frac{1}{n+1}\left[f(x)\right]^{n+1} + c \quad \text{where } n \neq -1$$

$$\int f'(x)\sin f(x)\,dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln\left|f(x)\right| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - \left[f(x)\right]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + \left[f(x)\right]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2n}\left\{f(a) + f(b) + 2\left[f\left(x_1\right) + \cdots + f\left(x_{n-1}\right)\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^{n}P_{r} = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

$$(x+a)^{n} = x^{n} + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^{r} + \cdots + a^{n}$$

Vectors

$$\left| \underset{\sim}{u} \right| = \left| x\underset{\sim}{i} + y\underset{\sim}{j} \right| = \sqrt{x^{2}+y^{2}}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = \left| \underset{\sim}{u} \right| \left| \underset{\sim}{v} \right| \cos\theta = x_{1}x_{2} + y_{1}y_{2},$$

where $\underset{\sim}{u} = x_{1}\underset{\sim}{i} + y_{1}\underset{\sim}{j}$

and $\underset{\sim}{v} = x_{2}\underset{\sim}{i} + y_{2}\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta)$$
$$= re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^{n} = r^{n}(\cos n\theta + i\sin n\theta)$$
$$= r^{n}e^{in\theta}$$

Mechanics

$$\frac{d^{2}x}{dt^{2}} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^{2}\right)$$

$$x = a\cos(nt + \alpha) + c$$

$$x = a\sin(nt + \alpha) + c$$

$$\ddot{x} = -n^{2}(x - c)$$

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